

Data Flow Diagram

Date	15 June 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	Farmer, Agricultural Expert, Weather Service – These entities provide agricultural and environmental data to the OptiCrop system and receive crop recommendations.
Rectangle with numbered header	Process	Collect Crop Data, Analyze Soil & Weather Conditions, Generate Crop Recommendations, Optimize Agricultural Production – These processes transform input data into useful recommendations and insights.
Rectangle (solid fill, no number)	Data Store	Crop Database, Soil Data Repository, Weather Data Repository, Recommendation Database – Stores agricultural, environmental, and recommendation data for future use.
Labeled arrow	Data Flow	Soil Nutrient Data (N, P, K), Temperature, Humidity, pH, Rainfall, Crop Information, Analysis Results, Crop Recommendations, Yield Optimization Suggestions – Represents movement of data between entities, processes, and data stores.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

